

DEV_IMPL_GUIDE - FOUNDATION_KAFKA

1. Purpose

This guide explains how to implement the Kafka foundation.

Use it with FOUNDATION_KAFKA.md.

This is a provisioning/runtime-setup guide. It is intentionally written as infrastructure setup steps rather than a business request execution flow.

2. What You Are Building

You are deploying Kafka for SOPHIX event transport:

- 4 Kafka nodes.
- KRaft mode.
- All nodes as brokers.
- 3 nodes as controllers.
- Topic naming, retention, replication, monitoring, and client rules.

3. Step 1 - Provision Nodes

Recommended medium setup:

Node	Broker	Controller
kafka-1	yes	yes
kafka-2	yes	yes
kafka-3	yes	yes
kafka-4	yes	no

Use separate disks for Kafka data where possible.

4. Step 2 - Configure Kafka

Set:

```
process.roles=broker,controller
controller.quorum.voters=1@kafka-1:9093,2@kafka-2:9093,3@kafka-3:9093
listeners=PLAINTEXT://:9092,CONTROLLER://:9093
log.dirs=/var/lib/kafka/data
num.partitions=6
default.replication.factor=3
min.insync.replicas=2
```

For kafka-4, use broker role only.

5. Step 3 - Create Core Topics

Create topics explicitly. Do not rely on auto-create in production.

Example:

```
kafka-topics --bootstrap-server kafka-1:9092 \
--create \
--topic sophix.subscription.operation.started \
--partitions 6 \
--replication-factor 3
```

Topic rules:

- Use `sophix.<domain>.<entity>.<event>`.
- Use `subscription_id` as key for subscription events.
- Use replication factor 3.

- Use `min.insync.replicas=2`.

6. Step 4 - Configure Producers

Recommended producer settings:

```
acks=all
enable.idempotence=true
retries=2147483647
max.in.flight.requests.per.connection=5
delivery.timeout.ms=120000
```

Application rule:

- Domain services write to `kafka_outbox`.
- Outbox dispatcher publishes to Kafka.
- Do not publish directly inside business DB transactions.

7. Step 5 - Configure Consumers

Consumer rules:

- Consumers must be idempotent.
- Store processed event IDs where needed.
- Use stable consumer group names.
- Do not assume exactly-once business behavior from Kafka alone.

8. Step 6 - Monitoring

Alert on:

Metric	Alert
Broker down	Immediate.
Controller quorum issue	Immediate.
Under-replicated partitions	Above zero.
Offline partitions	Above zero.
Consumer lag	Above SLA threshold.
Disk usage	Above 80 percent.

9. AWS Option

If using AWS:

- Prefer Amazon MSK for managed Kafka.
- Use private subnets.
- Enable encryption in transit and at rest.
- Use CloudWatch/OpenMonitoring.
- Keep replication factor 3.

10. Tests

Test:

- Create topic.
- Produce message.
- Consume message.
- Stop one broker and confirm cluster remains available.

- Confirm under-replicated partitions recover.
- Confirm outbox dispatcher can publish.

11. Common Mistakes

- Using one partition for high-volume events.
- Using random keys and losing per-subscription order.
- Setting replication factor 1.
- Letting auto-created topics into production.
- Treating Kafka publish as a replacement for DB transaction safety.